

Chapter 2

Inner product and norms

Hereafter, we will always work with vectors in $\mathbb{F}^n$, and matrices in $\mathbb{F}^{m \times n}$ ($m \leq n$), where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

Mathematically, a *vector norm* for vectors in the vector space $V = \mathbb{F}^n$ is a non-negative, positively homogeneous function $\|\cdot\|$, satisfying the triangular inequality. That is, it must satisfy these properties, for any pair of vectors $v, w \in V$ and any scalar $\alpha \in \mathbb{F}$:

$$\begin{cases} \|v\| \geq 0, & \text{and } \|v\| = 0 \text{ only if } v = 0, \\ \|\alpha v\| = |\alpha| \|v\|, \\ \|v + w\| \leq \|v\| + \|w\|. \end{cases} \quad (2.0.1)$$

When $V = \mathbb{F}^{m \times n}$, the norm is also defined as above and it is called *matrix norm*. We further say that the matrix norm is *consistent* (or *submultiplicative*) with respect to matrix product if, whenever the product below is well defined, one has:

$$\|vz\| \leq \|v\| \|z\|. \quad (2.0.2)$$

[Note that in (2.0.2), we can have $v \in \mathbb{F}^{m \times p}$, $z \in \mathbb{F}^{p \times n}$].

Some matrix norms are *induced by vector norms*. Namely, for $A \in \mathbb{F}^{m \times n}$, these are defined as

$$\|A\| = \max_{v \in \mathbb{F}^n: \|v\|=1} \|Av\|, \quad (2.0.3)$$

and induced norms are consistent.

2.1 Inner Product

Let us recall the operational definition of the Euclidean (or standard) inner product $\langle x, y \rangle$ for vectors x, y in $\mathbb{C}^n$ (or $\mathbb{R}^n$):

$$x, y \in \mathbb{C}^n \Rightarrow \langle x, y \rangle = y^* x ,$$

$$x, y \in \mathbb{R}^n \Rightarrow \langle x, y \rangle = y^T x ,$$

where we are using the notation: $y^* = \bar{y}^T$ (conjugate transpose, also called Hermitian of y).

Note that in $\mathbb{R}^n$ we have $\langle x, y \rangle = \langle y, x \rangle$, but the same is not true in $\mathbb{C}^n$, in general. In fact, in $\mathbb{R}^n$, the inner product is bilinear, that is for any $\alpha, \beta \in \mathbb{R}$:

- a) $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$;
- b) $\langle x, \alpha y + \beta z \rangle = \alpha \langle x, y \rangle + \beta \langle x, z \rangle$;

In $\mathbb{C}^n$, the inner product is linear in the first argument, but “conjugate-linear” in the second:

- c) $\langle x, \alpha y + \beta z \rangle = \bar{\alpha} \langle x, y \rangle + \bar{\beta} \langle x, z \rangle$, for any $\alpha, \beta \in \mathbb{C}$. In particular, $\langle x, y \rangle = \overline{\langle y, x \rangle}$.

Regardless of being in $\mathbb{C}$ or $\mathbb{R}$, the norm of a vector x (Euclidean norm, 2-norm, or simply length) induced by the above inner product is

$$\|x\|_2 = (\langle x, x \rangle)^{1/2} . \quad (2.1.1)$$

Likewise, the induced matrix norm (or just 2-norm, also called the *operator norm*) of $A \in \mathbb{F}^{m \times n}$ is

$$\|A\|_2 = \max_{x \in \mathbb{F}^n: \|x\|_2=1} \|Ax\|_2 . \quad (2.1.2)$$

A most important result is the following inequality.

Lemma 2.1.1 (Cauchy-Schwartz inequality) *For any $x, y \in \mathbb{F}^n$, we have*

$$|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2 , \quad (2.1.3)$$

with equality only if $y = \alpha x$.

Exercise 2.1.2 *Prove Lemma 2.1.1. [Hint: Consider the quadratic polynomial $g(t) = \langle x + ty, x + ty \rangle$, for $t \in \mathbb{R}$.]*

Exercise 2.1.3 *Use (2.1.3) to deduce the triangular inequality:*

$$\|x + y\|_2 \leq \|x\|_2 + \|y\|_2 . \quad (2.1.4)$$

Remark 2.1.4 In $\mathbb{R}^n$, an immediate consequence of the Cauchy-Schwartz inequality is that we can define an angle θ between x and $y \in \mathbb{R}^n$ from the relation:

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\|_2 \|y\|_2}, \quad (2.1.5)$$

customarily taking $0 \leq \theta \leq \pi$.

In $\mathbb{C}^n$, the consensus is to define the angle θ between $x = x_r + ix_i$ and $y = y_r + iy_i \in \mathbb{C}^n$ from the relation:

$$\cos \theta = \frac{\operatorname{Re}(\langle x, y \rangle)}{\|x\|_2 \|y\|_2}, \quad (2.1.6)$$

which is just (2.1.5) for the angle between the vectors $\begin{bmatrix} x_r \\ x_i \end{bmatrix}$ and $\begin{bmatrix} y_r \\ y_i \end{bmatrix}$ in $\mathbb{R}^{2n}$.

Once we have the concept of inner product, it is natural to define perpendicularity, orthogonality, of vectors when their inner product is 0 (which implies that the angle θ is $\pi/2$).

Definition 2.1.5 Two vectors (in $\mathbb{R}^n$ or $\mathbb{C}^n$) are called orthogonal if their inner product is 0. Further, we call them orthonormal if each of them has unit length.

We know (and it is also a consequence of the Gram-Schmidt process below) that there is an orthonormal basis for $\mathbb{F}^n$ over $\mathbb{F}$ (e.g., just consider the standard unit vectors $\{e_i\}$, $i = 1, \dots, n$). In particular, given an orthonormal basis $\{u_k\}_{k=1}^n$, any vector v can be uniquely written as

$$v = \alpha_1 u_1 + \dots + \alpha_n u_n, \quad \text{and} \quad \alpha_k = \langle v, u_k \rangle, \quad k = 1, \dots, n.$$

The values α_k are the *components* of v along the direction of u_k , $k = 1, \dots, n$.

A very commonly employed process is the “Gram-Schmidt orthogonalization,” which allows us to transform a given set of p linearly independent vectors into another set of p orthonormal vectors, spanning the same subspace as the original set.

Algorithm 2.1.6 (Gram-Schmidt) Given $\{x_1, x_2, \dots, x_p\}$ linearly independent vectors in $\mathbb{C}^n$ (or $\mathbb{R}^n$). The procedure below will produce $\{u_1, u_2, \dots, u_p\}$ such that $\langle u_i, u_j \rangle = 0$, $i \neq j$, $\langle u_i, u_i \rangle = 1$, and $\operatorname{span}\{u_1, u_2, \dots, u_k\} = \operatorname{span}\{x_1, x_2, \dots, x_k\}$, for $k = 1, 2, \dots, p$.

$i = 1 :$

$$y_1 = x_1 \quad , \quad u_1 = y_1 / \|y_1\|_2 \quad .$$

$i > 1 :$

$$\begin{cases} y_2 = x_2 - \langle x_2, u_1 \rangle u_1 & (\text{subtract from } x_2 \text{ its component in the direction of } u_1) \\ u_2 = y_2 / \|y_2\|_2 & (\text{normalize}) \\ \vdots \\ y_i = x_i - \langle x_i, u_{i-1} \rangle u_{i-1} - \cdots - \langle x_i, u_1 \rangle u_1 & , \quad i = 2, \dots, p. \\ u_i = y_i / \|y_i\|_2 \end{cases}$$

From the above algorithm, we can rewrite the original vectors as follows:

$$\begin{aligned} x_1 &= u_1 \|y_1\|_2 \quad , \\ x_2 &= u_1 \langle x_2, u_1 \rangle + u_2 \|y_2\|_2 \quad , \\ x_3 &= u_1 \langle x_3, u_1 \rangle + u_2 \langle x_3, u_2 \rangle + u_3 \|y_3\|_2 \quad , \\ &\vdots \\ x_p &= u_1 \langle x_p, u_1 \rangle + u_2 \langle x_p, u_2 \rangle + \cdots + u_{p-1} \langle x_p, u_{p-1} \rangle + u_p \|y_p\|_2 \quad . \end{aligned}$$

Therefore, if we write $U = [u_1, u_2, \dots, u_p] \in \mathbb{C}^{n \times p}$ and $X = [x_1, x_2, \dots, x_p] \in \mathbb{C}^{n \times p}$, then we have found that $X = UR$, where $R \in \mathbb{C}^{p \times p}$ is upper triangular. Moreover, the diagonal of R is positive (it is $\|y_k\|_2$, $k = 1, \dots, p$). Finally, the matrix U has *orthonormal* columns: $u_j^* u_k = 0$, $j \neq k$, and $u_j^* u_j = 1$, $j = 1, \dots, p$. For this reason, the matrix U is called *orthonormal*: $U^* U = I_p$. If $p = n \Rightarrow U$ is called *unitary*: $U^* U = I$ ($= U U^*$).

Also, note that the Gram-Schmidt process holds unchanged in the real case. In this case, when $p = n$, U is called *orthogonal*: $U^T U = I$ ($= U U^T$). [We will reserve the term orthogonal for real matrices (some texts use orthogonal also for $A \in \mathbb{C}^{n \times n}$ whenever $A^T A = I$).]

Exercise 2.1.7 Give the formula for the non-zero entries of R obtained from the Gram-Schmidt process.

We summarize, and extend, the construction given by the Gram-Schmidt process in the following theorem.

Theorem 2.1.8 (QR factorization) *Let $A \in \mathbb{C}^{n \times p}$ (or $\mathbb{R}^{n \times p}$), $n \geq p$, be full rank. Then there is an orthonormal (n, p) matrix Q and a (p, p) upper triangular and invertible R , with real diagonal, such that $A = QR$. This factorization is unique for any assigned sign pattern on the diagonal of R (e.g., all positive). If $p = n$, then Q is unitary (orthogonal in the real case). $\square$*

Exercise 2.1.9 *Verify Theorem 2.1.8. (Note that the only thing you need to do is to verify that one can tweak the Gram-Schmidt process to obtain any desired sign pattern on the diagonal of R).*

The following are important properties of unitary, respectively orthogonal, matrices U .

- (1) $U^{-1} = U^*$ and U^* is unitary (respectively, orthogonal).
- (2) UV is unitary (orthogonal) if U and V are.
- (3) With respect to standard matrix multiplication, the set $S = \{U \in \mathbb{C}^{n \times n}, U \text{ unitary}\}$ is a group. It is a subgroup of $GL(n, \mathbb{C})$, the general linear group: all invertible (n, n) matrices with entries in $\mathbb{C}$; for this reason, the set S is often written as $GU(n, \mathbb{C})$. Similarly, the orthogonal matrices are a subgroup of $GL(n, \mathbb{R})$, often written as $GO(n, \mathbb{R})$.
[Recall that a group is a set G with an operation, write it “.”, such that: (i) the set G is closed with respect to “.”, (ii) the operation “.” is associative, (iii) there is an identity element with respect to “.”, and (iv) there is an inverse (with respect to “.”) for every element of G . A subgroup is a subset S of G , which is itself a group with respect to the operation “.”.]
- (4) The set $\{U \in \mathbb{C}^{n \times n}, U \text{ unitary}\}$ is closed and bounded (as subset of $\mathbb{C}^{n^2}$) and therefore compact. Likewise, the set $\{Q \in \mathbb{R}^{n \times n}, Q \text{ orthogonal}\}$ is closed and bounded (as subset of $\mathbb{R}^{n^2}$) and therefore compact.
- (5) $\|Ux\|_2 = \|x\|_2, \forall x \in \mathbb{C}^n$, and thus $\|U\|_2 = 1$. Similarly, if Q is orthogonal, then $\|Qx\|_2 = \|x\|_2, \forall x \in \mathbb{R}^n$, and thus $\|Q\|_2 = 1$. [Multiplication by a unitary (orthogonal) matrix maintains length].

Exercises 2.1.10

- (1) *Show properties (1), (2), (3), (4), (5) above. [Hint: To show property (4), reason as follows. Let $\{U_k\}$ be a sequence of unitary matrices and suppose that $U_k \rightarrow U$. Show that U is unitary. Here, convergence is entrywise.]*

- (2) Show that all the eigenvalues of a unitary matrix U are on the unit circle, so that $|\det U| = 1$. The result is true also in the real case, but $\det U$ can only take values ± 1 in the real case. [As an aside remark, this means that there are two disjoint classes of orthogonal matrices: Those with determinant 1 and those with determinant -1 . In the plane, these are *rotations* and *reflections*, respectively.]
- (3) Suppose that $A \in \mathbb{C}^{n \times n}$ is similar to U , and that U is unitary. Show that A^{-1} is similar to A^* .
- (4) Show that the angle between any two vectors x and y in $\mathbb{C}^n$ is maintained if the vectors are multiplied by a unitary matrix.

Let us now see some consequences of *unitary similarity* (orthogonal similarity in $\mathbb{R}^{n \times n}$). By this we mean that matrices A and B are related as $A = U^*BU$, where U is unitary.

First of all, observe that unitary similarity is an equivalence relation, and that the 2-norm of a matrix is unitarily invariant, since unitary transformations maintain length; that is, if U is unitary

$$\|A\|_2 = \|U^*AU\|_2.$$

An important consequence of this is that unitary similarity is a “stable” transformation, in the sense that one does not magnify perturbations:

“If E : $\|E\|_2 \leq \eta$, then $\|U^*(A + E)U\|_2 \leq \|U^*AU\|_2 + \eta$.”

Another matrix norm: Frobenius norm

The 2-norm of a matrix is not the only unitarily invariant norm¹. A commonly used norm is the *Frobenius norm* $\|A\|_F$ of a matrix $A \in \mathbb{F}^{m \times n}$:

$$\|A\|_F^2 = \sum_{i,j} |a_{ij}|^2. \quad (2.1.7)$$

This is also called the Hilbert-Schmidt norm, and it satisfies

$$\|A\|_F^2 = \text{tr}(A^*A). \quad (2.1.8)$$

Note that $\|A\| = \|\text{vec}(A)\|_2$, where $\text{vec}(A) = \begin{bmatrix} A(:,1) \\ A(:,2) \\ \vdots \\ A(:,n) \end{bmatrix} \in \mathbb{F}^{mn}$.

¹More examples are in Remark 5.1.23 and Exercise 7.5.2

Exercise 2.1.11 Show that the Frobenius norm is in fact a norm. Also, verify (2.1.8).

Theorem 2.1.12 (Frobenius norm invariance) Let $A \in \mathbb{C}^{m \times n}$, let $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ be unitary, and let $B = U^*AV$. Then

$$\|B\|_F = \|A\|_F.$$

Similarly, if $A \in \mathbb{R}^{m \times n}$, U, V orthogonal, and $B = Q^T AZ$.

Pf. As we saw, $\sum_{i,j} |a_{ij}|^2 = \text{tr}(A^*A)$. But $\text{tr}(A^*A) = \text{tr}(V^*A^*UU^*AV) = \text{tr}((U^*AV)^*(U^*AV))$ since the trace is invariant under similarity transformations (see Remark 1.5.59). Therefore, $\text{tr}(A^*A) = \text{tr}(B^*B)$. Similarly in the real case. $\square$

Remark 2.1.13 Careful: Similar matrices may fail to satisfy Frobenius norm invariance. For example, take $A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -9 & -110 \\ 1 & 12 \end{bmatrix}$, $\left(B = T^{-1}AT, T = \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix} \right)$.

Exercise 2.1.14 Let $A \in \mathbb{C}^{n \times n}$ (or also in $\mathbb{R}^{n \times n}$). Show that $\|A\|_2 \leq \|A\|_F$.

2.2 Other norms

There are many other norms for vectors (and matrices) which are routinely used. We recall a few important ones. We can think of vectors in $\mathbb{C}^n$ and matrices in $\mathbb{C}^{m \times n}$.

1-norm $\|x\|_1 = \sum_i |x_i|;$

p-norms $\|x\|_p = (\sum_i |x_i|^p)^{1/p}$, for $1 < p < \infty$;

sup-norm (also called infinity norm) $\|x\|_\infty = \max_i |x_i|.$

The induced matrix norms for all above $p = 1, \dots, \infty$, are just

$$\|A\|_p = \max_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} = \max_{\|x\|_p=1} \|Ax\|_p.$$

Remark 2.2.1 *Not all matrix norms are induced. For example, the Frobenius norm is not an induced matrix norm. Neither is the max-element norm*

$$\|A\|_{\max} := \max_{i,j} |a_{ij}| . \quad (2.2.1)$$

Exercises 2.2.2

- (i) *Verify that $\|A\|_{\max}$ in (2.2.1) is a matrix norm.*
- (ii) *Show that all matrix p -norms ($p = 1, 2, \dots, \infty$) are consistent (that is, they satisfy (2.0.2)), and that so is the Frobenius norm. Is $\|A\|_{\max}$ a consistent matrix norm? Show it, or give a counterexample.*

Luckily, there are operational formulas which make the evaluation of some matrix norms a rather simple thing. For one thing, $\|A\|_F$ is not hard to compute, but also the following formulas are quite useful for practical computations:

$$\begin{aligned} \|A\|_{\infty} &= \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}| \\ \|A\|_1 &= \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}| . \end{aligned} \quad (2.2.2)$$

Also, recall that –in finite dimension– all **norms are equivalent**. That is, given an n -dimensional vector space V , and any two norms, say $\|\cdot\|_{\alpha}$ and $\|\cdot\|_{\beta}$, then there are positive real constants b and c (that may depend on n) such that

$$b\|v\|_{\beta} \leq \|v\|_{\alpha} \leq c\|v\|_{\beta} , \quad \forall v \in V . \quad (2.2.3)$$

Exercise 2.2.3 *Find sharp constants b and c , such that:*

- a) $b\|v\|_{\infty} \leq \|v\|_2 \leq c\|v\|_{\infty} , \quad v \in \mathbb{R}^n ,$
- b) $b\|v\|_1 \leq \|v\|_{\infty} \leq c\|v\|_1 , \quad v \in \mathbb{C}^n .$

Our next exercise introduces an important quantity, the spectral radius of a matrix, which will play a key role in Chapter 4.

Exercise 2.2.4 (On Spectral Radius) Let $A \in \mathbb{C}^{n \times n}$. By definition, the spectral radius of A is the quantity

$$\rho(A) := \max_{1 \leq i \leq n} |\lambda_i|, \quad (2.2.4)$$

where $\lambda_1, \dots, \lambda_n$, are the eigenvalues of A .

- (i) By definition, the matrix A is called convergent if $\lim_{k \rightarrow \infty} A^k = 0$ (that is, all entries of A^k go to 0 in the limit of $k \rightarrow \infty$). Show that A is convergent if and only if $\rho(A) < 1$.

- (ii) Show that

$$\rho(A) = \lim_{k \rightarrow \infty} \|A^k\|^{1/k},$$

where $\|\cdot\|$ is any consistent matrix norm. As an after thought, argue that the result is true for any norm, whether consistent or not (because all norms in $\mathbb{C}^{n \times n}$ are equivalent).

- (iii) By giving a counterexample, show that the spectral radius of a matrix is not a matrix norm, in general.

Exercise 2.2.5 (Weighted norm)

- (i) Let $\|\cdot\|$ be a norm on the n -dimensional vector space $\mathbb{F}^n$ and let $A \in \mathbb{F}^{n \times n}$ be an invertible matrix. Consider the following quantity:

$$\|v\|_A = \|Av\|, \quad \forall v \in \mathbb{F}^n. \quad (2.2.5)$$

Show that (2.2.5) is a vector norm. Since often A is a diagonal matrix, the norm (2.2.5) is also called a weighted norm.

- (ii) Similarly, if $\|\cdot\|$ is a matrix norm on $\mathbb{F}^{m \times n}$ and $A \in \mathbb{F}^{m \times m}$ and $B \in \mathbb{F}^{n \times n}$ are invertible matrices, then show that the following is a matrix norm:

$$\|C\|_{A,B} = \|ACB\|, \quad \forall C \in \mathbb{F}^{m \times n}. \quad (2.2.6)$$

Exercise 2.2.6 Let $A \in \mathbb{C}^{n \times n}$. Using Exercises 2.2.4 and 2.2.5, as well as Exercise 7.1.4, show that, although $\rho(A)$ is not a matrix norm, for any given $\varepsilon > 0$, there is a consistent matrix norm $\|\cdot\|$, such that $\rho(A) \leq \|A\| \leq \rho(A) + \varepsilon$.

